DELEX by Piradex — measurements

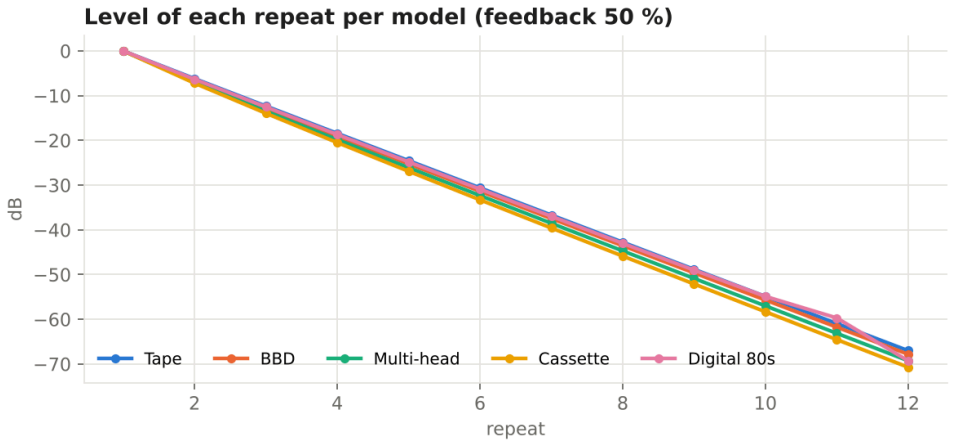
Objective measurements of the DELEX 1.0 engine, following the **ANALYSER by Piradex** methodology: impulse and short noise for each repeat's response, sines for THD, harmonics and aliasing, a level sweep for the ducker, silence for the noise floor. All at 48 kHz, single precision, as in the plugin. The numbers come from `tests/measure.cpp` and the curves from `tools/plot_medicoes.py` — anyone can repeat them.

Summary

Measurement	Result
Tape: drop per repeat (feedback 50 %)	-6.1 dB
1st repeat: 60 Hz relative to 1 kHz	-3.2 dB
1st repeat: 14 kHz relative to 1 kHz	-3.0 dB
6th repeat: 10 kHz relative to 1 kHz	-3.0 dB
THD at DRIVE 60 % (1 kHz, −6 dBFS)	17.2 %
Alias at 2.5 kHz (normal)	-45.4 dB
Alias at 2.5 kHz (HQ 2x)	-109.5 dB
Noise floor (silence, feedback 90 %)	digital silence
HISS at maximum	-55.2 dBFS RMS
Max echo time error (1 ms … 8 s)	1 samples
Mono sum of out-of-phase echoes, GUARD off	-150.0 dB
Mono sum of out-of-phase echoes, GUARD on	-0.0 dB
Base CPU (one core, 48 kHz stereo)	2.2 %
CPU with everything on	8.8 %
Plugin latency	0 samples

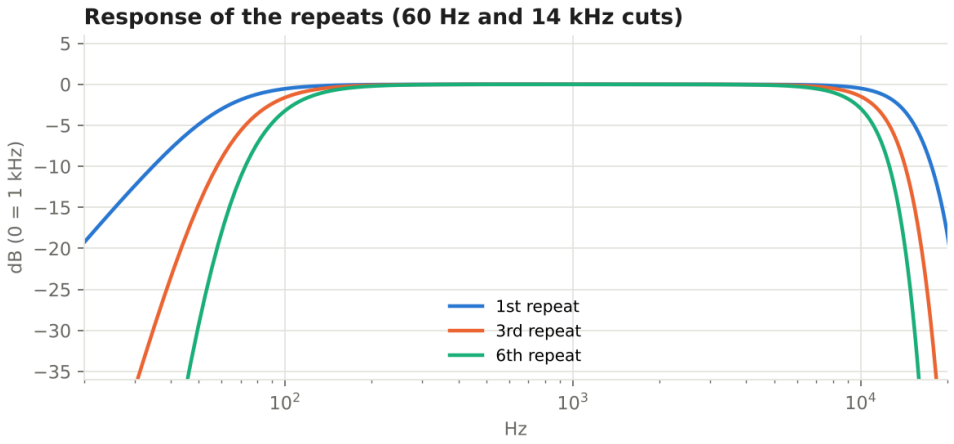
Echo tail per model

At 50 % feedback each repeat drops 6 dB (the maths says 20·log10(0.5) = −6.02 dB): FEEDBACK does exactly what it says. Cassette loses a little more because its narrow band and saturation act on every pass.



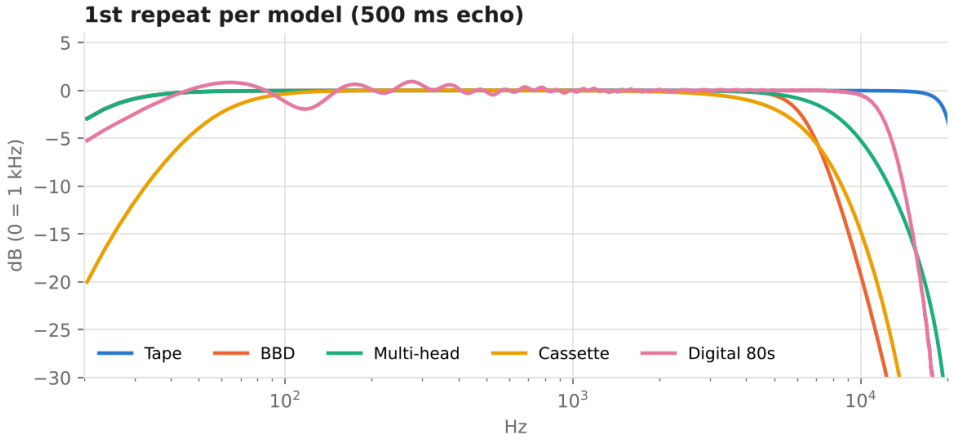
Response of the repeats

Low cut 60 Hz and high cut 14 kHz (12 dB/octave) inside the loop. The cuts add up on every pass: repeats get progressively darker and lighter in the lows, like a well-tuned analog delay.



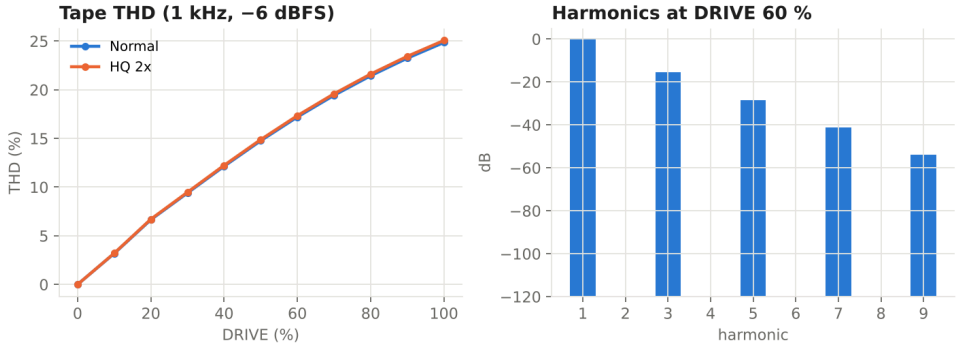
Character of the models

First repeat of each model, normalised at 1 kHz.



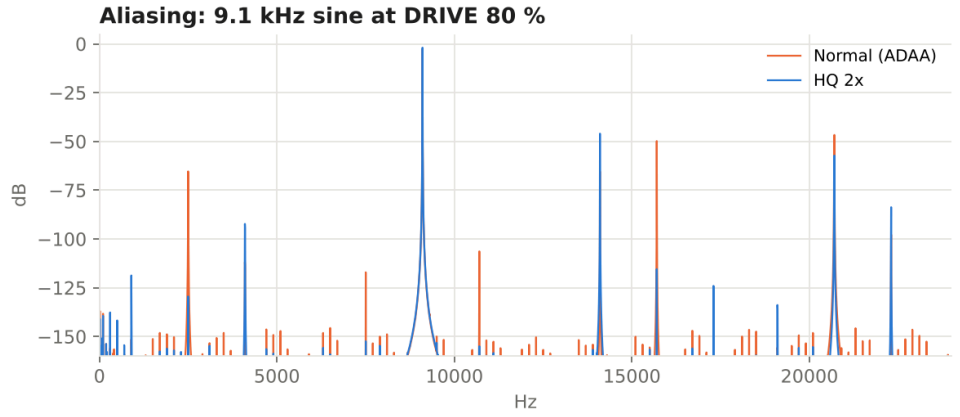
Tape saturation

Distortion rises smoothly with DRIVE and is dominated by low odd harmonics (tape character). At 0 % DRIVE the loop is linear.



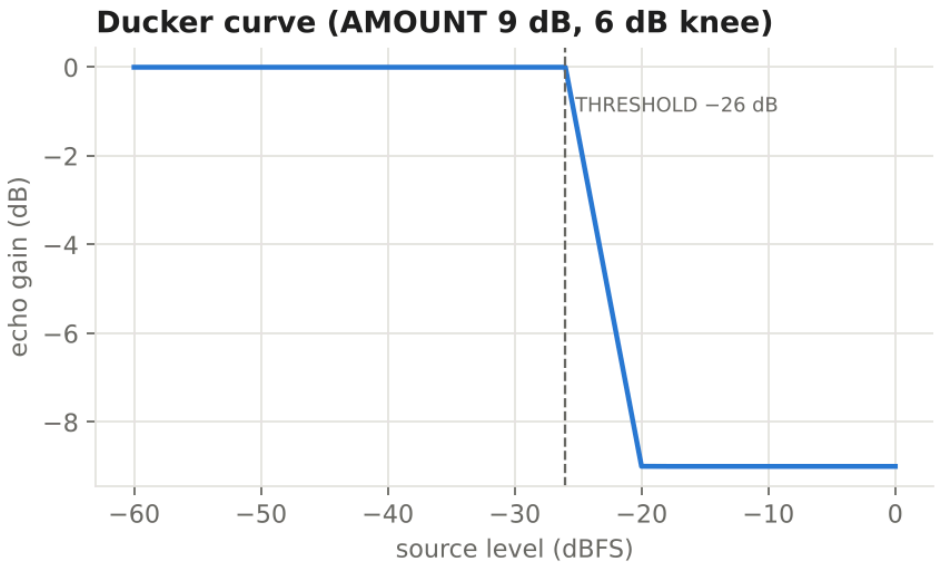
Aliasing

Even without oversampling the saturation uses antiderivative anti-aliasing (ADAA). HQ 2x pushes aliasing below the noise floor.



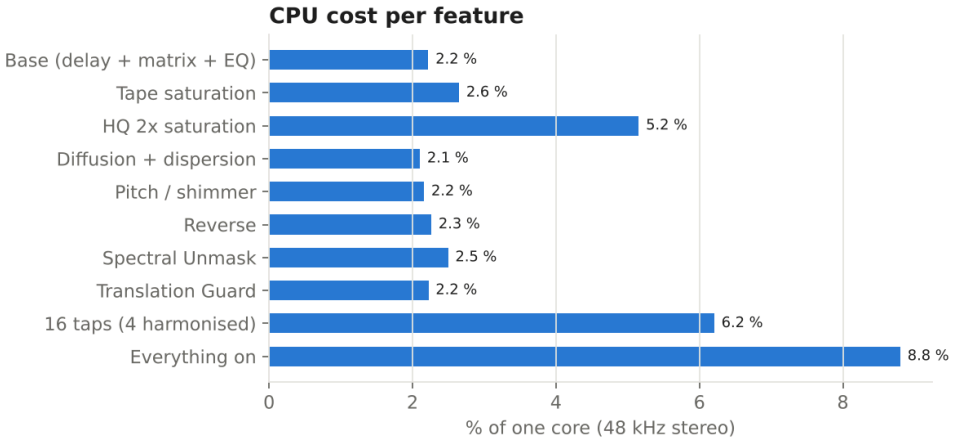
Ducker

Reduction versus source level, with a soft 6 dB knee from the THRESHOLD.



CPU

Measured on a modern core; a 2013 MacBook Pro uses roughly twice as much, still comfortable.



Repeat: `g++ -O2 -std=c++17 -ISource tests/measure.cpp -o measure && ./measure docs/medicoes/data && python3 tools/plot_medicoes.py`